

# An Amortized Continuous PSF Field for DES Single-Epoch Imaging

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## ABSTRACT

Accurate point-spread-function (PSF) models are the dominant systematic in weak-lensing shear measurement. The standard approach fits an independent PSF model per exposure (e.g., PSFEX, PIFF) from that exposure’s stars. We instead learn a single *amortized* continuous PSF field: a multi-head attention network takes an exposure’s stars as context (position, color, flux, and pixel cutouts) and predicts the PSF at any queried position and resolution, with no per-exposure fitting at test time. On 3,301 frozen test exposures of Dark Energy Survey (DES) single-epoch imaging (CCD 31, *grizY*; 73,609 reserved stars), the model matches or exceeds PIFF and PSFEX on every metric: size-residual scatter 0.030 versus 0.039/0.037, ellipticity errors  $|\delta e_1|, |\delta e_2| = 0.015/0.015$  versus 0.017/0.016 (PIFF), and shape correlations 0.65/0.61 versus 0.61/0.58 (PIFF), at  $\chi^2/\text{dof} = 1.04$ . In a star-anchored galaxy injection–recovery test on real data, our PSF recovers galaxy *ellipticity* far better than PIFF ( $\Delta e_1 = +0.004$  versus  $-0.138$ ), the weak-lensing-critical quantity. Color conditioning removes a chromatic PSF-size systematic that the baselines retain ( $+0.004/\text{mag}$  versus  $\sim -0.06/\text{mag}$  residual slope). The model is flat in the number of available fit stars where PIFF degrades sharply ( $\chi^2 \simeq 1.1$  versus 5.1 at five stars). Using simulations as a controlled instrument, we show that a *blend forward-model* training loss removes a size bias from unrecognized blends, and we identify and fix a spin-2 representation failure in which the network corrupts the PSF ellipticity by attending to galaxy detections in its context; restricting attention to point sources recovers it fully.

**Keywords:** Weak gravitational lensing (1797) — Point spread function (1349) — Astronomical techniques (1684) — Convolutional neural networks (1938) — Sky surveys (1464)

## 1. INTRODUCTION

Weak gravitational lensing infers the matter distribution from coherent  $\sim 1\%$  distortions of galaxy shapes. The measurement is limited by knowledge of the PSF: any error in the modeled PSF size or ellipticity propagates directly into the inferred shear. Survey pipelines model the PSF per exposure by fitting that exposure’s stars with a flexible model interpolated across the focal plane (PSFEX; E. Bertin 2011) (PIFF; M. Jarvis et al. 2021). This is robust but (i) discards information shared across exposures and nights, (ii) degrades when few clean stars are available, and (iii) is purely empirical at the star positions—it does not exploit auxiliary information such as stellar color, which sets the chromatic PSF.

We take an amortized, learning-based approach. A single network is trained across many exposures to map an exposure’s stars (the *context*) to a continuous PSF field, queryable at arbitrary position, color, flux, and

resolution. The inferential target is the PSF at *star-free* positions, where galaxies are measured; predicting held-out stars is the training objective and the validation instrument. We train and evaluate on DES single-epoch data and benchmark against PIFF and PSFEX run by us under matched conditions.

## 2. METHOD

*Architecture*—Each detected source is encoded from its pixel cutout (a learned image encoder applied to the scale-normalized stamp), its sky position (sinusoidal encoding), and scalar color and log-flux. A multi-head self-attention block lets every query attend to the context sources; a coordinate decoder then renders the PSF at the query position and any oversampling factor, returning the pixel-convolved effective PSF used to convolve galaxy models. We use a polar (spin-2-aware) angular parametrization of the decoder and FiLM modulation of the decoder by the attended context features.

*Loss*—Training minimizes an inverse-variance  $\chi^2$  between predicted and observed star cutouts, with the per-star amplitude solved in closed form. We introduce a *blend forward-model* loss: rather than treat each clean star as isolated, we model its cutout as a generalized-least-squares sum of the target plus its detected neighbors, all rendered by the same decoder (§4.5). Detected galaxies near a target are handled by exclusion (default), masking, or a crude extended-component model; the three are within noise on real reserved stars.

*Point-source context*—The PSF should be informed only by point sources. We restrict the attention context to stellar detections; §4.5 shows this is necessary to avoid a spin-2 corruption from extended sources in the context.

*Color conditioning*—The decoder conditions on  $g-i$  color, enabling a per-object chromatic PSF (differential chromatic refraction and wavelength-dependent seeing) that per-exposure empirical models cannot represent (§4.3).

### 3. DATA

We use DES single-epoch FITS (Y5A1, CCD 31; amplifier B is dead, masked via MSK bit 8). Sources are detected with SEP and cross-matched to DES DR2 for colors, `SPREAD_MODEL` star/galaxy labels, and coadd object IDs. Splits are at the observing-night level by deterministic hash to prevent leakage (consecutive exposures share atmosphere), frozen in a committed manifest with  $\sim 20\%$  of clean stars per test/validation exposure reserved for scoring only; the test split is 3,301 exposures across *grizY*. Reserved stars are excluded from every method’s fit/context and scored identically across methods on the same pixel grid.

*Simulations*—A matched pipeline renders Moffat PSF fields (second-order polynomial FWHM and shear variation—exactly representable by PIFF/PSFEX, a deliberately fair-to-baselines choice) into the same data format, optionally with chromatic FWHM–color dependence, blended stars, and injected galaxy detections. Star density, galaxy density (3:1), and per-source S/N are tuned to match the real extraction. The simulation is the controlled instrument for the systematics studies in §4.5.

## 4. RESULTS

### 4.1. Reserved-star accuracy on real data

On 3,301 test exposures and 73,609 reserved stars, the amortized model (no per-exposure fitting) matches or exceeds both baselines on every metric (Table 1). Reserved “clean” stars carry a mild blend-contamination

floor that suppresses absolute correlations for *all* methods; the comparison is fair (identical stars) and the ordering is robust.

### 4.2. Galaxy recovery on real data

We test the inferential goal directly. We inject GALSIM galaxies, convolved with a strictly isolated reserved star’s empirical PSF, at that star’s position; each method predicts the PSF there from the *other* stars and recovers the galaxy. The truth arm (the anchor’s own image) recovers near-zero bias, validating the protocol. On 113 *r*-band exposures and 522 anchors (Table 2), our PSF recovers galaxy *ellipticity* essentially without bias, whereas PIFF’s PSF-ellipticity error induces a large shape bias ( $\Delta e_1 = -0.14$ ). PIFF recovers size slightly better; our PSF is  $\sim 6\%$  too broad at star-free positions. For weak lensing, the ellipticity result is the consequential one.

### 4.3. Chromatic PSF correction

In a chromatic simulation (FWHM varies  $-3\%/mag$  of  $g-i$ ), color conditioning reduces the residual chromatic size slope from  $-0.06/mag$  (PIFF, PSFEX, and our own zero-color ablation) to  $+0.004/mag$ —a  $\sim 17\times$  reduction. On real *r*-band data the paired color-versus-zero-color slope difference is  $+0.0024/mag$  (CI  $[+0.0016, +0.0030]$ ); the baselines structurally retain the systematic. This is information unavailable to a per-exposure empirical PSF.

### 4.4. Sample efficiency

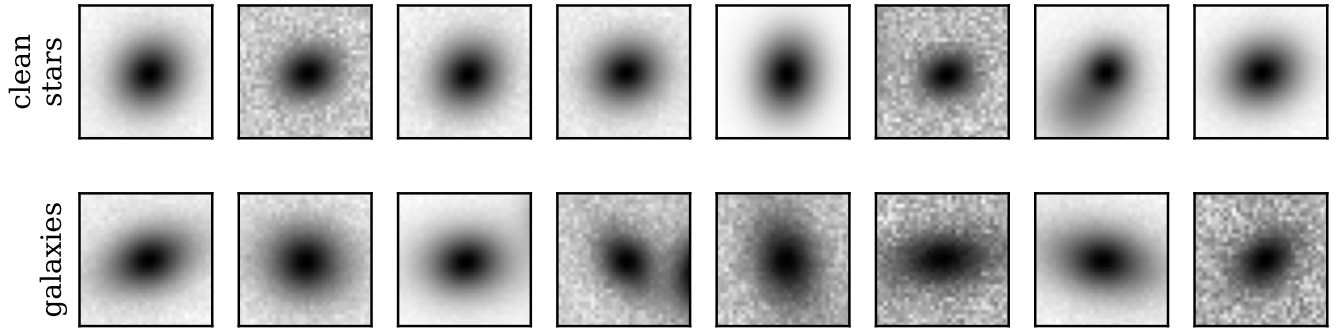
Restricting every method to  $k$  randomly chosen fit stars, the amortized model is approximately flat in  $k$  (size scatter  $0.078 \rightarrow 0.056$ ,  $\chi^2 \sim 1.1$  from  $k = 5$  to  $k = 100$ ), while PIFF collapses below  $k \approx 25$  ( $\chi^2 = 5.1$ , scatter  $0.18$  at  $k = 5$ ). Amortization transfers information from the training set to sparsely-populated exposures.

### 4.5. Simulation studies

*Architecture ladder*—On the star-free truth grid,  $\chi^2/dof$  improves along additive  $\rightarrow$  FiLM  $\rightarrow$  diagonal  $\rightarrow$  polar ( $8.9 \rightarrow 4.96 \rightarrow 4.41$ ; PIFF 2.67; verified floor 1.0), recovering both ellipticity components.

*Blend forward-model*—On a blended simulation, single-star training leaves a  $+4.8\%$  size bias on the star-free truth grid; the blend forward-model loss reduces it to  $+0.7\%$  ( $\approx 7\times$ ).

*Galaxy-context corrupts PSF ellipticity—and the fix*—Injecting galaxy detections into the simulation, a model that attends to *all* detections (including galaxies) suffers a clean, reproducible failure: it cannot represent

Example simulated cutouts ( $32 \times 32$  px, arcsinh stretch)

**Figure 1.** Example simulated cutouts. Clean stars (top) are compact point sources; injected galaxies (bottom) are visibly extended and elliptical (second-moment size  $T \approx 33$  vs  $15 \text{ px}^2$ ). Both are presented to the network as context.

**Table 1.** Reserved-star residuals on the DES test split (3,301 exposures, 73,609 stars).  $T$  scatter is the robust standard deviation of  $\delta T/T$ ;  $|\delta e|$  are median absolute ellipticity residuals; corr is the model–star shape correlation. Best in bold.

| Method           | $T$ scat      | $ \delta e_1 $ | $ \delta e_2 $ | corr( $e_1$ ) | corr( $e_2$ ) | $\chi^2/\text{dof}$ |
|------------------|---------------|----------------|----------------|---------------|---------------|---------------------|
| <b>This work</b> | <b>0.0301</b> | <b>0.0151</b>  | <b>0.0150</b>  | <b>0.650</b>  | <b>0.607</b>  | 1.037               |
| PIFF             | 0.0393        | 0.0171         | 0.0158         | 0.611         | 0.577         | 1.073               |
| PSFEx            | 0.0371        | 0.0160         | 0.0154         | 0.645         | 0.604         | 1.047               |

**Table 2.** Star-anchored galaxy injection–recovery (113  $r$ -band exposures, 522 anchors). The truth arm validates the protocol.

| Arm                | Size bias | $\Delta e_1$  | $\Delta e_2$ |
|--------------------|-----------|---------------|--------------|
| Truth (validation) | −1.5%     | −0.002        | $\sim 0$     |
| <b>This work</b>   | −7.6%     | <b>+0.004</b> | −0.002       |
| PIFF               | −1.0%     | <b>−0.138</b> | −0.001       |

one spin-2 ellipticity component [ $\text{corr}(e_1) = 0.05$  under the polar parametrization; the failure flips to  $e_2$  under a Cartesian parametrization, identifying a representation degeneracy rather than noise], while recovering the other perfectly. We localize the cause by ablation: it is *not* data volume, per-source S/N, galaxy size or color, or the  $\chi^2$  cap—the model is treating extended galaxy cutouts as PSF evidence. Restricting the attention context to point sources recovers *both* components completely [ $\text{corr}(e_1, e_2) = 0.98$ , matching the galaxy-free ideal]; a coordinate-system change does not. Galaxies are clearly distinguishable (cutout second mo-

ment  $T \approx 33$  versus  $15 \text{ px}^2$  for stars), so this is the model failing to use available information, fixed by the principled restriction that only point sources inform a PSF.

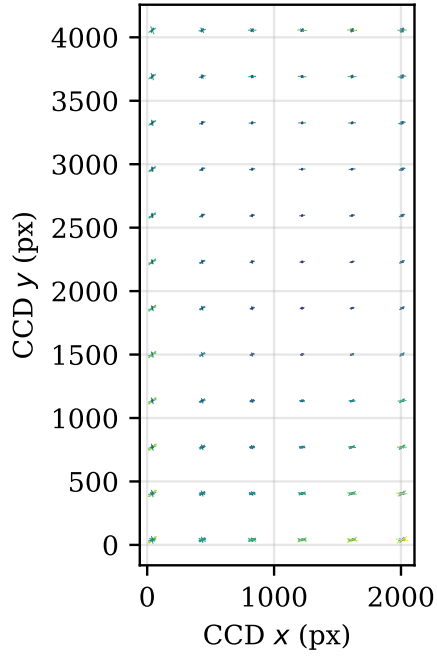
## 5. DISCUSSION

The amortized field trades a per-exposure fit for a single trained model. Benefits: no test-time fitting, robustness in star-sparse exposures, and conditioning on color (chromatic PSF) and flux (brighter–fatter). Caveats: (i) unlike per-exposure fitters, the amortized model requires many exposures to learn the field (in simulation, ellipticity recovery is unstable below a few thousand exposures; real DES is well above this); (ii) the simulation’s idealized PSF (Moffat, low-order polynomial field) favors the per-exposure baselines, so absolute-accuracy comparisons rest on real data; and (iii) the galaxy-context finding argues that a PSF estimator’s context should be gated to point sources, which we adopt.

## 6. CONCLUSION

A single attention-based continuous PSF field, trained across DES single-epoch exposures and applied with no per-exposure fitting, matches or exceeds PIFF and

True PSF ellipticity field (simulation)



**Figure 2.** The true, spatially varying PSF ellipticity field in a simulated exposure (whisker length  $\propto |e|$ ); the network must recover this at star-free positions.

PSFEX on real reserved stars and recovers galaxy ellipticity better than PIFF on a star-anchored injection test, while additionally correcting a chromatic systematic and remaining robust in star-sparse exposures. Controlled simulations motivate two training choices—a blend forward-model loss and point-source-only attention context—the latter from a clean diagnosis of a spin-2 representation failure triggered by galaxy detections in context.

#### ACKNOWLEDGMENTS, FUNDING, DES CONSORTIUM STATEMENT.

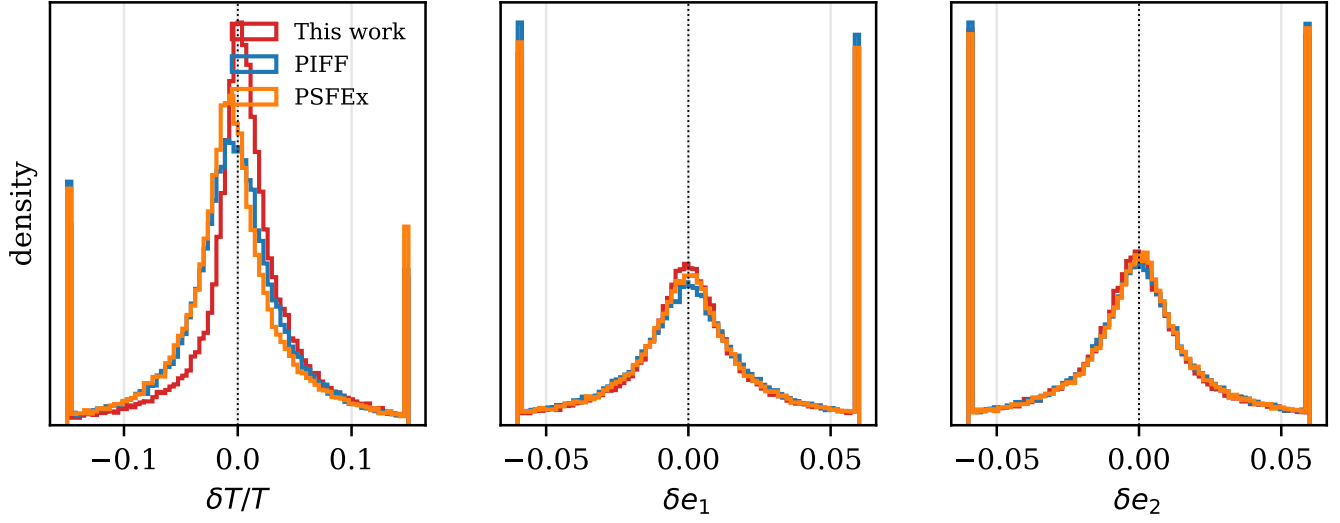
Baselines: PIFF (PixelGrid, second-order BasisPolynomial,  $4\sigma$  rejection) and PSFEX (PSF\_SAMPLING 0.5,  $45 \times 45$ , PSFVAR\_DEGREES 2; rendered via GALSIM DES\_PSFEX, method='no\_pixel').

*Software:* Astropy (Astropy Collaboration 2022), GalSim (B. T. P. Rowe et al. 2015), SEP (K. Barbary 2016), PyTorch (A. Paszke et al. 2019), PIFF (M. Jarvis et al. 2021), PSFEX (E. Bertin 2011)

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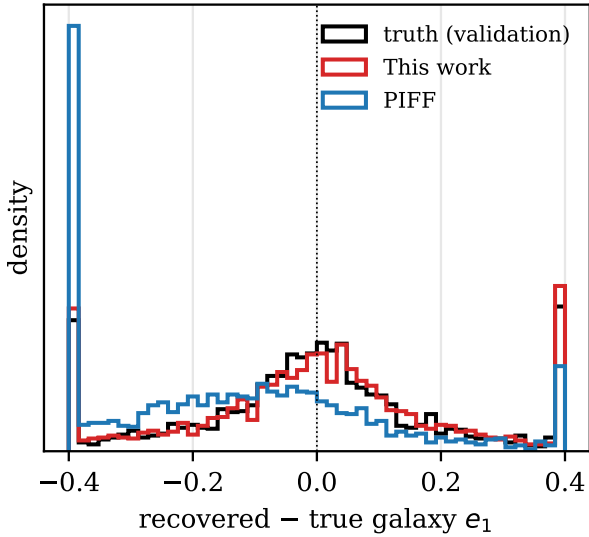
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## Reserved-star residuals on the DES test split

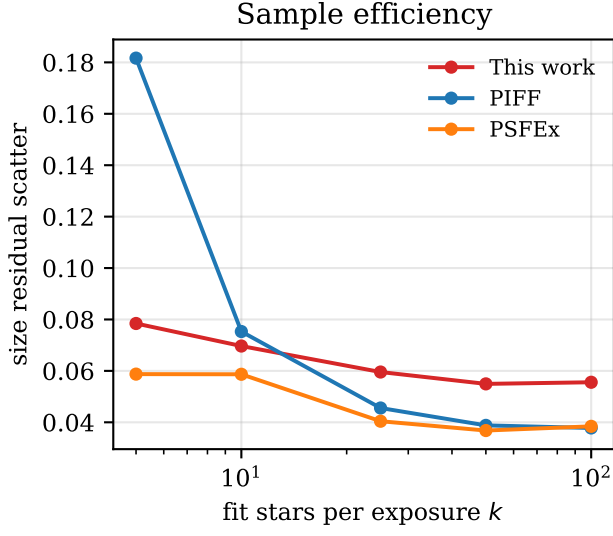


**Figure 3.** Reserved-star residual distributions on the DES test split. Our amortized model (red) is the narrowest in size and ellipticity.

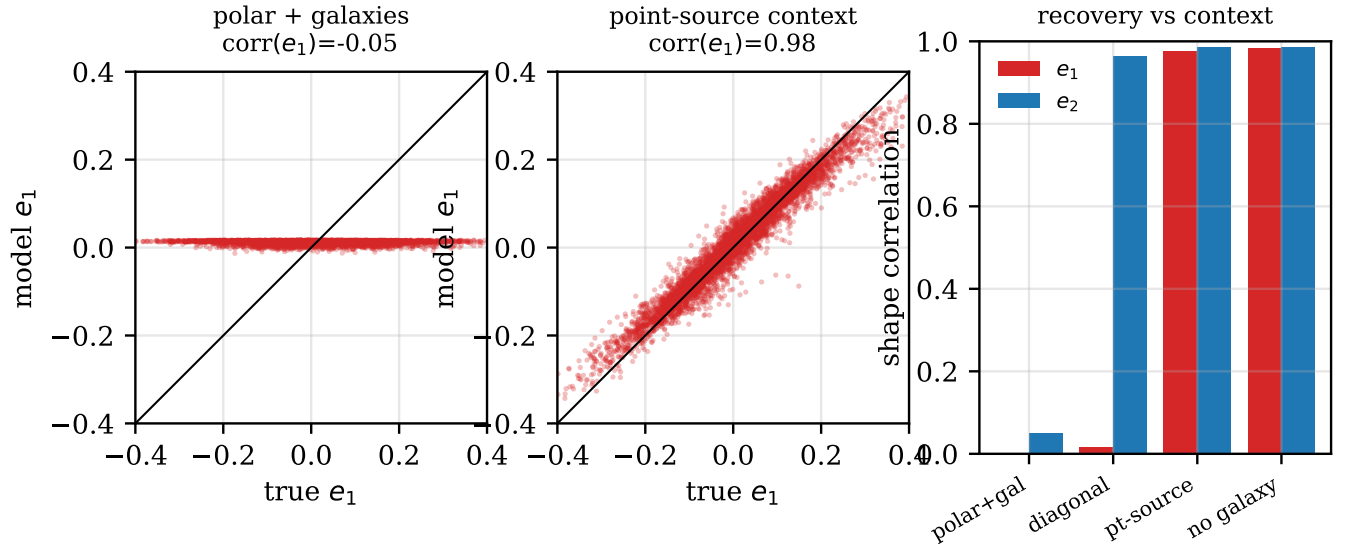
## Galaxy ellipticity recovery (real data)



**Figure 4.** Galaxy ellipticity recovery on real data (star-anchored injection). PIFF's PSF-ellipticity error biases recovered galaxy  $e_1$  by  $-0.14$ ; ours is unbiased.



**Figure 5.** Size-residual scatter versus the number of fit stars  $k$ . The amortized model is nearly flat in  $k$ ; PIFF degrades sharply below  $k \approx 25$ .



**Figure 6.** Galaxy detections in the attention context corrupt the recovered PSF ellipticity. Left: with galaxies in context, model  $e_1$  collapses to a constant ( $\text{corr} = -0.04$ ). Middle: restricting the context to point sources recovers it ( $\text{corr} = 0.98$ ). Right: shape correlations across configurations—point-source context and galaxy-free both recover *both* components; a coordinate change (diagonal) does not.